LAPORAN PRAKTIKUM PERANCANGAN JARINGAN

Pemilihan dan Penerapan Protokol Switching dan Routing #3



Disusun oleh

Nama	:	Azura Rizky Annisa
NIM	:	24/535715/SV/24290
Hari, Tanggal	:	Jum’at. 24 April 2026
Kelas	:	RI4B1

PROGRAM STUDI DIPLOMA IV TEKNOLOGI REKAYASA INTERNET
DEPARTEMEN TEKNIK ELEKTRO DAN INFORMATIKA SEKOLAH VOKASI
UNIVERSITAS GADJAH MADA

2026

TUJUAN

a. Memahami perancangan jaringan secara *top-down*
b. Menerapkan rancangan jaringan berhirarki
c. Melakukan konfigurasi protokol-protokol
   - VLAN
   - Subnetting dan Pengalamatan IP
   - Inter-VLAN Router
   - DHCP
   - SSH
   - RIPv2
   - Switching Security
   - Host
d. Pengujian dan Verifikasi komunikasi jaringan

DASAR TEORI

Faktor-faktor yang dilibatkan dalam sebuah pengambilan keputusan :

- Tujuan harus jelas
- Pilihan-pilihan atau ide yang ada harus dipelajari lebih jauh terlebih dahulu.
- Konsekuensi dari setiap keputusan harus dipelajari lebih lanjut
- Rencana cadangan harus segera dibuat.

Selain faktor-faktor diatas juga harus melibatkan beberapa pertanyaan berikut ini :

- Jika hasil keputusan yang diambil ternyata salah, apa yang akan dikerjakan ?
- Jika hasil keputusan sudah pernah dilakukan sebelumnya, apakah ada permasalahan dengan keputusan tersebut ?
- Bagaimana reaksi pelanggan dengan keputusan anda ?
- Apa rencana cadangan anda jika keputusan anda ditolak oleh pelanggan ?

Berikut ini adalah salah satu contoh dokumen dalam pembuatan keputusan

	Critical Goals			Other Goals		
	Adaptability—must adapt to changes in a large internetwork within seconds	Must scale to a large size (hundreds of routers)	Must be an industry standard and compatible with existing equipment	Should not create a lot of traffic	Should run on inexpensive routers	Should be easy to configure and manage
BGP	X*	X	X	8	7	7
OSPF	X	X	X	8	8	8
IS-IS	X	X	X	8	6	6
IGRP	X	X				
EIGRP	X	X				
RIP			X			

*X = Meets critical criteria. 1 = Lowest. 10 = Highest.

Pemilihan Switching Protocols

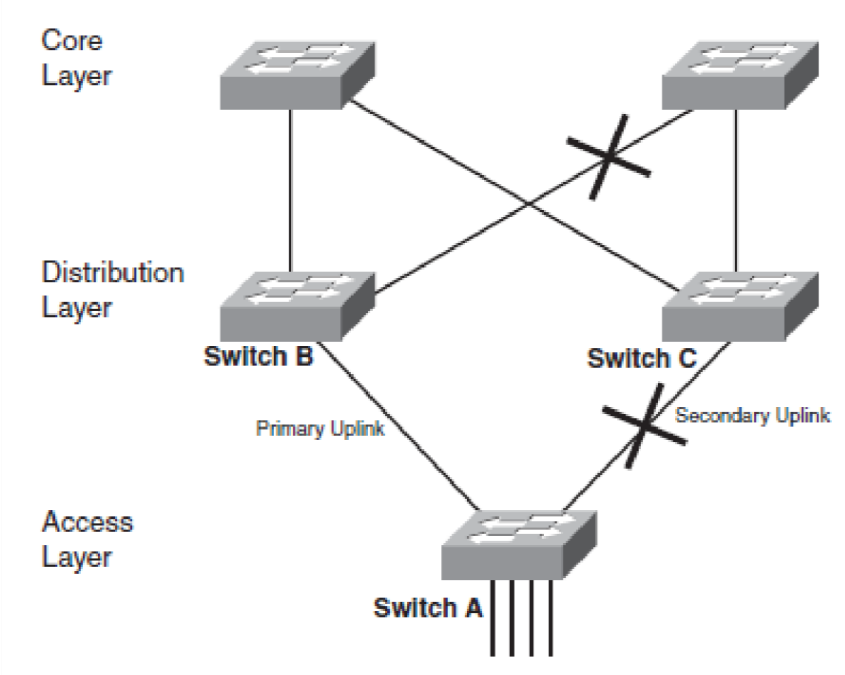
Switch memanfaatkan sirkuit terintegrasi cepat untuk menawarkan latensi rendah. Bridges jauh lebih lambat daripada switch dan memiliki lebih sedikit port dan biaya per port yang lebih tinggi. Untuk alasan ini, switch telah menggantikan bridges. Switch memiliki kemampuan untuk melakukan pemrosesan *store-and-forward* atau pemrosesan *cut-through*. Dengan pemrosesan *cut-through*, sebuah switch dengan cepat melihat alamat tujuan (bidang pertama dalam header frame Ethernet), menentukan port keluar, dan segera mulai mengirim bit ke port keluar. Kerugian dengan pemrosesan *cut-through* adalah meneruskan frame ilegal (misalnya, kerdil Ethernet) dan frame dengan kesalahan CRC.

Pada jaringan yang rawan kegagalan dan kesalahan, pemrosesan *cut-through* tidak boleh digunakan. Beberapa switch memiliki kemampuan untuk secara otomatis berpindah dari mode *cut-through* ke mode *store-and forward* ketika ambang kesalahan tercapai. Fitur ini disebut pengalihan *cut-through adaptif* oleh beberapa vendor. Switch memungkinkan beberapa jalur penerusan paralel, yang berarti sebuah switch dapat menangani volume lalu lintas yang tinggi lebih cepat daripada sebuah bridge. Switch high-end dapat mendukung banyak jalur penerusan simultan, tergantung pada struktur switching.

Berikut ini beberapa protokol switching yang dapat digunakan :

- STP
- RSTP
- PVST
- PVST+
- Etherchannel
- VTP
- DTP

Berikut ini contoh penggunaan STP



Gambar 1 Switch lapisan akses dengan dua uplinks ke switches layer distribusi

Pemilihan Routing Protocols

Protokol routing memungkinkan router secara dinamis mempelajari cara menjangkau jaringan lain dan bertukar informasi ini dengan router atau host lain. Memilih protokol routing untuk pelanggan desain jaringan Anda lebih sulit daripada memilih protokol switching, karena ada begitu banyak pilihan. Keputusan akan lebih mudah jika Anda dapat menggunakan tabel keputusan, seperti yang ditunjukkan pada Tabel 7-1 pada buku *Top-Down Network Design 3rd Edition.* Berbekal pemahaman yang kuat tentang tujuan dan informasi pelanggan Anda tentang karakteristik protokol routing yang berbeda, Anda dapat membuat keputusan yang tepat tentang protokol routing mana yang direkomendasikan. Berikut ini tabel ringkasan panduan dalam pemilihan protocol routing

Tabel 1 Perbandingan protokol-protokol routing

	Distance Vector or Link State	Interior or Exterior	Classful or Classless	Metrics Supported	Scalability	Convergence Time	Resource Consumption	Supports Security? Authenticates Routes?	Ease of Design, Configuration, and Troubleshooting
RIPv1	Distance vector	Interior	Classful	Hop count	15 hops	Can be long (if no load balancing)	Memory: low CPU: low Bandwidth: high	No	Easy
RIPv2	Distance vector	Interior	Classless	Hop count	15 hops	Can be long (if no load balancing)	Memory: low CPU: low Bandwidth: high	Yes	Easy
IGRP	Distance vector	Interior	Classful	Bandwidth, delay, reliability, load	255 hops (default is 100)	Quick (uses triggered updates and poison reverse)	Memory: low CPU: low Bandwidth: high	No	Easy
EIGRP	Advanced distance vector	Interior	Classless	Bandwidth, delay, reliability, load	1000s of routers	Very quick (uses DUAL algorithm)	Memory: moderate CPU: low Bandwidth: low	Yes	Easy
OSPF	Link state	Interior	Classless	Cost (100 million divided by bandwidth on Cisco routers)	A few hundred routers per area, a few hundred areas	Quick (uses LSAs and Hello packets)	Memory: high CPU: high Bandwidth: low	Yes	Moderate
BGP	Path vector	Exterior	Classless	Value of path attributes and other configurable factors	1000s of routers	Quick (uses update and keepalive packets, and withdraws routes)	Memory: high CPU: high Bandwidth: low	Yes	Moderate
IS-IS	Link state	Interior	Classless	Configured path value, plus delay, expense, and errors	Hundreds of routers per area, a few hundred areas	Quick (uses LSAs)	Memory: high CPU: high Bandwidth: low	Yes	Moderate

Berikut juga adalah tabel administrative distance berdasarkan tipe route untuk preferensi penentuan rute oleh router.

Tabel 2 Administrative Distance berdasarkan tipe route

Route Source	Default Distance Value
Connected interface	0
Static route	1
Enhanced Interior Gateway Routing Protocol (EIGRP) summary route	5
External Border Gateway Protocol (BGP)	20
Internal EIGRP	90
IGRP	100
OSPF	110
Intermediate System-to-Intermediate System (IS-IS)	115
Routing Information Protocol (RIP)	120
Exterior Gateway Protocol (EGP)	140
On-Demand Routing (ODR)	160
External EIGRP	170
Internal BGP	200
Unknown	255

Simpulan Pemilihan Protokol

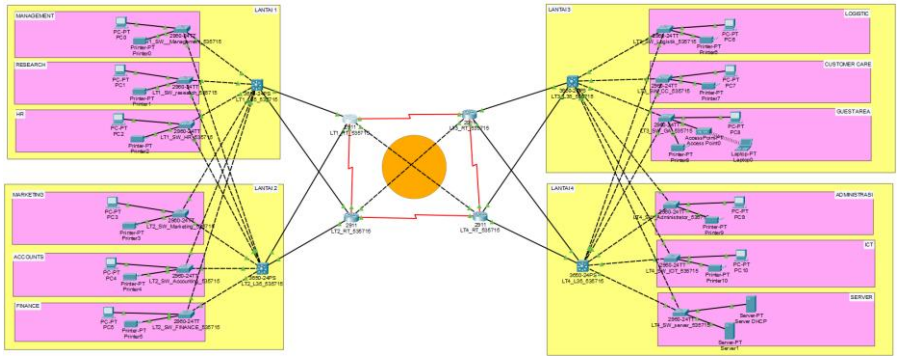
Pemilihan protokol switching dan routing yang tepat untuk pelanggan desain jaringan Anda sangat diperlukan. Skalabilitas dan karakteristik kinerja protokol dan seberapa cepat protokol dapat beradaptasi dengan perubahan menjadi perhatian penting dalam pemilihan protocol witching maupun routing.

Memutuskan protokol switching dan routing yang tepat untuk pelanggan Anda akan membantu Anda memilih produk switch dan router terbaik untuk pelanggan. Misalnya, jika Anda telah memutuskan bahwa desain harus mendukung protokol routing yang dapat menyatu dalam hitungan detik di internetwork besar, Anda mungkin tidak akan merekomendasikan router yang hanya menjalankan RIP.

Detail pembahasan bagaimana memilih protocol switching dan routing sangaat direkomendasikan untuk membaca buku Top-down Network Design 3rd Edition dari CISCO yang juga sebagai buku utama matakuliah teori Perancangan Jaringan. Buku tersebut terdapat diskusi umum tentang pengambilan keputusan untuk membantu Anda mengembangkan proses sistematis untuk memilih solusi desain jaringan. Sebuah diskusi tentang menjembatani dan beralih protokol diikuti, yang meliputi menjembatani transparan, beralih multilayer, perangkat tambahan untuk protokol STP, dan VLAN. Bagian pada protokol routing mengikuti bagian switching.

HASIL PRAKTIKUM

Topologi



Checklist Pengujian Konfigurasi Topologi

1. Cek interface pada tiap perangkat. Screenshot pada masing-masing L3 Switch dan router
    - L3S Lantai 1

```
LT1_L35_535715>en
Password:
LT1_L35_535715#show ip int brief
Interface              IP-Address      OK? Method Status                Protocol
GigabitEthernet1/0/1   10.10.10.17     YES manual up                    up
GigabitEthernet1/0/2   10.10.10.33     YES manual up                    up
GigabitEthernet1/0/3   unassigned      YES unset  up                    up
GigabitEthernet1/0/4   unassigned      YES unset  up                    up
GigabitEthernet1/0/5   unassigned      YES unset  up                    up
GigabitEthernet1/0/6   unassigned      YES unset  up                    up
GigabitEthernet1/0/7   unassigned      YES unset  up                    up
GigabitEthernet1/0/8   unassigned      YES unset  up                    up
GigabitEthernet1/0/9   unassigned      YES unset  down                  down
GigabitEthernet1/0/10  unassigned      YES unset  down                  down
GigabitEthernet1/0/11  unassigned      YES unset  down                  down
GigabitEthernet1/0/12  unassigned      YES unset  down                  down
GigabitEthernet1/0/13  unassigned      YES unset  down                  down
GigabitEthernet1/0/14  unassigned      YES unset  down                  down
GigabitEthernet1/0/15  unassigned      YES unset  down                  down
GigabitEthernet1/0/16  unassigned      YES unset  down                  down
GigabitEthernet1/0/17  unassigned      YES unset  down                  down
GigabitEthernet1/0/18  unassigned      YES unset  down                  down
GigabitEthernet1/0/19  unassigned      YES unset  down                  down
GigabitEthernet1/0/20  unassigned      YES unset  down                  down
GigabitEthernet1/0/21  unassigned      YES unset  down                  down
GigabitEthernet1/0/22  unassigned      YES unset  down                  down
GigabitEthernet1/0/23  unassigned      YES unset  down                  down
GigabitEthernet1/0/24  unassigned      YES unset  down                  down
GigabitEthernet1/1/1   unassigned      YES unset  down                  down
GigabitEthernet1/1/2   unassigned      YES unset  down                  down
GigabitEthernet1/1/3   unassigned      YES unset  down                  down
GigabitEthernet1/1/4   unassigned      YES unset  down                  down
Vlan1                  unassigned      YES unset  administratively down down
Vlan10                 192.168.10.1    YES manual up                    up
Vlan20                 192.168.10.65   YES manual up                    up
Vlan30                 192.168.10.129  YES manual up                    up
Vlan40                 192.168.10.193  YES manual up                    up
Vlan50                 192.168.11.1    YES manual up                    up
Vlan60                 192.168.11.65   YES manual up                    up
LT1_L35_535715#
```

    - L3S Lantai 2

```
LT2_L35_535715>en
Password:
LT2_L35_535715#show ip int brief
Interface              IP-Address      OK? Method Status                Protocol
GigabitEthernet1/0/1   10.10.10.29     YES manual up                    up
GigabitEthernet1/0/2   10.10.10.37     YES manual up                    up
GigabitEthernet1/0/3   unassigned      YES unset  up                    up
GigabitEthernet1/0/4   unassigned      YES unset  up                    up
GigabitEthernet1/0/5   unassigned      YES unset  up                    up
GigabitEthernet1/0/6   unassigned      YES unset  up                    up
GigabitEthernet1/0/7   unassigned      YES unset  up                    up
GigabitEthernet1/0/8   unassigned      YES unset  up                    up
GigabitEthernet1/0/9   unassigned      YES unset  down                  down
GigabitEthernet1/0/10  unassigned      YES unset  down                  down
GigabitEthernet1/0/11  unassigned      YES unset  down                  down
GigabitEthernet1/0/12  unassigned      YES unset  down                  down
GigabitEthernet1/0/13  unassigned      YES unset  down                  down
GigabitEthernet1/0/14  unassigned      YES unset  down                  down
GigabitEthernet1/0/15  unassigned      YES unset  down                  down
GigabitEthernet1/0/16  unassigned      YES unset  down                  down
GigabitEthernet1/0/17  unassigned      YES unset  down                  down
GigabitEthernet1/0/18  unassigned      YES unset  down                  down
GigabitEthernet1/0/19  unassigned      YES unset  down                  down
GigabitEthernet1/0/20  unassigned      YES unset  down                  down
GigabitEthernet1/0/21  unassigned      YES unset  down                  down
GigabitEthernet1/0/22  unassigned      YES unset  down                  down
GigabitEthernet1/0/23  unassigned      YES unset  down                  down
GigabitEthernet1/0/24  unassigned      YES unset  down                  down
GigabitEthernet1/1/1   unassigned      YES unset  down                  down
GigabitEthernet1/1/2   unassigned      YES unset  down                  down
GigabitEthernet1/1/3   unassigned      YES unset  down                  down
GigabitEthernet1/1/4   unassigned      YES unset  down                  down
Vlan1                  unassigned      YES unset  administratively down down
Vlan10                 192.168.10.1    YES manual up                    up
Vlan20                 192.168.10.65   YES manual up                    up
Vlan30                 192.168.10.129  YES manual up                    up
Vlan40                 192.168.10.193  YES manual up                    up
Vlan50                 192.168.11.1    YES manual up                    up
Vlan60                 192.168.11.65   YES manual up                    up
LT2_L35_535715#
```

- L3S Lantai 3

```
LT3_L3S_535715>en
Password:
LT3_L3S_535715#show ip int brief
Interface              IP-Address      OK? Method Status                Protocol
GigabitEthernet1/0/1   10.10.10.21     YES manual up                    up
GigabitEthernet1/0/2   10.10.10.41     YES manual up                    up
GigabitEthernet1/0/3   unassigned      YES unset  up                    up
GigabitEthernet1/0/4   unassigned      YES unset  up                    up
GigabitEthernet1/0/5   unassigned      YES unset  up                    up
GigabitEthernet1/0/6   unassigned      YES unset  up                    up
GigabitEthernet1/0/7   unassigned      YES unset  up                    up
GigabitEthernet1/0/8   unassigned      YES unset  up                    up
GigabitEthernet1/0/9   unassigned      YES unset  down                  down
GigabitEthernet1/0/10  unassigned      YES unset  down                  down
GigabitEthernet1/0/11  unassigned      YES unset  down                  down
GigabitEthernet1/0/12  unassigned      YES unset  down                  down
GigabitEthernet1/0/13  unassigned      YES unset  down                  down
GigabitEthernet1/0/14  unassigned      YES unset  down                  down
GigabitEthernet1/0/15  unassigned      YES unset  down                  down
GigabitEthernet1/0/16  unassigned      YES unset  down                  down
GigabitEthernet1/0/17  unassigned      YES unset  down                  down
GigabitEthernet1/0/18  unassigned      YES unset  down                  down
GigabitEthernet1/0/19  unassigned      YES unset  down                  down
GigabitEthernet1/0/20  unassigned      YES unset  down                  down
GigabitEthernet1/0/21  unassigned      YES unset  down                  down
GigabitEthernet1/0/22  unassigned      YES unset  down                  down
GigabitEthernet1/0/23  unassigned      YES unset  down                  down
GigabitEthernet1/0/24  unassigned      YES unset  down                  down
GigabitEthernet1/1/1   unassigned      YES unset  down                  down
GigabitEthernet1/1/2   unassigned      YES unset  down                  down
GigabitEthernet1/1/3   unassigned      YES unset  down                  down
GigabitEthernet1/1/4   unassigned      YES unset  down                  down
Vlan1                  unassigned      YES unset  administratively down down
Vlan70                 192.168.11.129  YES manual up                    up
Vlan80                 192.168.11.193  YES manual up                    up
Vlan90                 192.168.12.1    YES manual up                    up
Vlan110                192.168.12.129  YES manual up                    up
Vlan120                192.168.12.193  YES manual up                    up
Vlan910                192.168.12.65   YES manual down                  down
LT3_L3S_535715#
```

- L3S Lantai 4

```
LT4_L3S_535715>en
Password:
LT4_L3S_535715#show ip int brief
Interface              IP-Address      OK? Method Status                Protocol
GigabitEthernet1/0/1   10.10.10.25     YES manual up                    up
GigabitEthernet1/0/2   10.10.10.45     YES manual up                    up
GigabitEthernet1/0/3   unassigned      YES unset  up                    up
GigabitEthernet1/0/4   unassigned      YES unset  up                    up
GigabitEthernet1/0/5   unassigned      YES unset  up                    up
GigabitEthernet1/0/6   unassigned      YES unset  up                    up
GigabitEthernet1/0/7   unassigned      YES unset  up                    up
GigabitEthernet1/0/8   unassigned      YES unset  up                    up
GigabitEthernet1/0/9   unassigned      YES unset  down                  down
GigabitEthernet1/0/10  unassigned      YES unset  down                  down
GigabitEthernet1/0/11  unassigned      YES unset  down                  down
GigabitEthernet1/0/12  unassigned      YES unset  down                  down
GigabitEthernet1/0/13  unassigned      YES unset  down                  down
GigabitEthernet1/0/14  unassigned      YES unset  down                  down
GigabitEthernet1/0/15  unassigned      YES unset  down                  down
GigabitEthernet1/0/16  unassigned      YES unset  down                  down
GigabitEthernet1/0/17  unassigned      YES unset  down                  down
GigabitEthernet1/0/18  unassigned      YES unset  down                  down
GigabitEthernet1/0/19  unassigned      YES unset  down                  down
GigabitEthernet1/0/20  unassigned      YES unset  down                  down
GigabitEthernet1/0/21  unassigned      YES unset  down                  down
GigabitEthernet1/0/22  unassigned      YES unset  down                  down
GigabitEthernet1/0/23  unassigned      YES unset  down                  down
GigabitEthernet1/0/24  unassigned      YES unset  down                  down
GigabitEthernet1/1/1   unassigned      YES unset  down                  down
GigabitEthernet1/1/2   unassigned      YES unset  down                  down
GigabitEthernet1/1/3   unassigned      YES unset  down                  down
GigabitEthernet1/1/4   unassigned      YES unset  down                  down
Vlan1                  unassigned      YES unset  administratively down down
Vlan70                 192.168.11.129  YES manual up                    up
Vlan80                 192.168.11.193  YES manual up                    up
Vlan90                 192.168.12.1    YES manual up                    up
Vlan100                192.168.12.65   YES manual up                    up
Vlan110                192.168.12.129  YES manual up                    up
Vlan120                192.168.12.193  YES manual up                    up
LT4 L3S 535715#
```

- RT 1

```
LT1_RT_535715>en
LT1_RT_535715#show ip int brief
Interface              IP-Address      OK? Method Status                Protocol
GigabitEthernet0/0     10.10.10.49     YES manual up                    up
GigabitEthernet0/1     10.10.10.18     YES manual up                    up
GigabitEthernet0/2     10.10.10.38     YES manual up                    up
Serial0/3/0            10.10.10.1      YES manual up                    up
Serial0/3/1            10.10.10.13     YES manual up                    up
Vlan1                  unassigned      YES unset  administratively down down
LT1 RT 535715#
```

- RT 2

```
LT2_RT_535715>en
LT2_RT_535715#show ip int brief
Interface              IP-Address      OK? Method Status                Protocol
GigabitEthernet0/0     10.10.10.53     YES manual up                    up
GigabitEthernet0/1     10.10.10.30     YES manual up                    up
GigabitEthernet0/2     10.10.10.34     YES manual up                    up
Serial0/3/0            10.10.10.14     YES manual up                    up
Serial0/3/1            10.10.10.9      YES manual up                    up
Vlan1                  unassigned      YES unset  administratively down down
LT2_RT_535715#
```

- RT 3

```
LT3_RT_535715>en
LT3_RT_535715#show ip int brief
Interface              IP-Address      OK? Method Status                Protocol
GigabitEthernet0/0     10.10.10.54     YES manual up                    up
GigabitEthernet0/1     10.10.10.22     YES manual up                    up
GigabitEthernet0/2     10.10.10.46     YES manual up                    up
Serial0/3/0            10.10.10.2      YES manual up                    up
Serial0/3/1            10.10.10.5      YES manual down                  down
Vlan1                  unassigned      YES unset  administratively down down
LT3_RT_535715#
```

- RT 4

```
LT4_RT_535715>en
LT4_RT_535715#show ip int brief
Interface              IP-Address      OK? Method Status                Protocol
GigabitEthernet0/0     10.10.10.50     YES manual up                    up
GigabitEthernet0/1     10.10.10.26     YES manual up                    up
GigabitEthernet0/2     10.10.10.42     YES manual up                    up
Serial0/3/0            10.10.10.10     YES manual up                    up
Serial0/3/1            10.10.10.6      YES manual down                  down
Vlan1                  unassigned      YES unset  administratively down down
LT4 RT 535715#
```

2. Cek routing pada tiap router dan L3 Switch.

- L3S Lantai 1

```
LT1_L35_535715#show ip route ospf
     10.0.0.0/8 is variably subnetted, 15 subnets, 2 masks
O       10.10.10.0 [110/66] via 10.10.10.34, 00:00:04, GigabitEthernet1/0/2
O       10.10.10.8 [110/65] via 10.10.10.34, 00:00:04, GigabitEthernet1/0/2
O       10.10.10.12 [110/65] via 10.10.10.34, 00:00:04, GigabitEthernet1/0/2
O       10.10.10.20 [110/3] via 10.10.10.34, 00:00:04, GigabitEthernet1/0/2
O       10.10.10.24 [110/5] via 10.10.10.34, 00:00:04, GigabitEthernet1/0/2
O       10.10.10.28 [110/2] via 10.10.10.34, 00:00:04, GigabitEthernet1/0/2
O       10.10.10.36 [110/3] via 10.10.10.34, 00:00:04, GigabitEthernet1/0/2
O       10.10.10.40 [110/4] via 10.10.10.34, 00:00:04, GigabitEthernet1/0/2
O       10.10.10.44 [110/3] via 10.10.10.34, 00:00:04, GigabitEthernet1/0/2
O       10.10.10.48 [110/5] via 10.10.10.34, 00:00:04, GigabitEthernet1/0/2
O       10.10.10.52 [110/2] via 10.10.10.34, 00:00:04, GigabitEthernet1/0/2
     192.168.12.0/26 is subnetted, 3 subnets
O       192.168.12.0 [110/4] via 10.10.10.34, 00:00:04, GigabitEthernet1/0/2
O       192.168.12.128 [110/4] via 10.10.10.34, 00:00:04, GigabitEthernet1/0/2
O       192.168.12.192 [110/4] via 10.10.10.34, 00:00:04, GigabitEthernet1/0/2

LT1_L35_535715#
```

- L3S Lantai 2

```
LT2_L35_535715#show ip route ospf
     10.0.0.0/8 is variably subnetted, 15 subnets, 2 masks
O       10.10.10.0 [110/66] via 10.10.10.30, 00:36:20, GigabitEthernet1/0/1
O       10.10.10.8 [110/65] via 10.10.10.30, 00:36:20, GigabitEthernet1/0/1
O       10.10.10.12 [110/65] via 10.10.10.30, 00:36:20, GigabitEthernet1/0/1
O       10.10.10.16 [110/3] via 10.10.10.30, 00:01:58, GigabitEthernet1/0/1
O       10.10.10.20 [110/3] via 10.10.10.30, 00:36:20, GigabitEthernet1/0/1
O       10.10.10.24 [110/5] via 10.10.10.30, 00:36:20, GigabitEthernet1/0/1
O       10.10.10.32 [110/2] via 10.10.10.30, 00:02:08, GigabitEthernet1/0/1
O       10.10.10.40 [110/4] via 10.10.10.30, 00:36:20, GigabitEthernet1/0/1
O       10.10.10.44 [110/3] via 10.10.10.30, 00:36:20, GigabitEthernet1/0/1
O       10.10.10.48 [110/5] via 10.10.10.30, 00:36:20, GigabitEthernet1/0/1
O       10.10.10.52 [110/2] via 10.10.10.30, 00:36:20, GigabitEthernet1/0/1
     192.168.12.0/26 is subnetted, 3 subnets
O       192.168.12.0 [110/4] via 10.10.10.30, 00:36:20, GigabitEthernet1/0/1
O       192.168.12.128 [110/4] via 10.10.10.30, 00:36:20, GigabitEthernet1/0/1
O       192.168.12.192 [110/4] via 10.10.10.30, 00:36:20, GigabitEthernet1/0/1

LT2_L35_535715#
```

- L3S Lantai 3

```
LT3_L35_535715#show ip route ospf
     10.0.0.0/8 is variably subnetted, 13 subnets, 2 masks
O       10.10.10.0 [110/65] via 10.10.10.22, 00:38:21, GigabitEthernet1/0/1
O       10.10.10.8 [110/65] via 10.10.10.42, 00:38:21, GigabitEthernet1/0/2
O       10.10.10.12 [110/66] via 10.10.10.42, 00:38:21, GigabitEthernet1/0/2
                    [110/66] via 10.10.10.22, 00:38:21, GigabitEthernet1/0/1
O       10.10.10.16 [110/3] via 10.10.10.42, 00:38:21, GigabitEthernet1/0/2
O       10.10.10.28 [110/3] via 10.10.10.22, 00:38:21, GigabitEthernet1/0/1
O       10.10.10.32 [110/3] via 10.10.10.22, 00:04:14, GigabitEthernet1/0/1
O       10.10.10.36 [110/3] via 10.10.10.42, 00:38:21, GigabitEthernet1/0/2
O       10.10.10.48 [110/2] via 10.10.10.42, 00:38:21, GigabitEthernet1/0/2
O       10.10.10.52 [110/2] via 10.10.10.22, 00:38:21, GigabitEthernet1/0/1
     192.168.10.0/26 is subnetted, 4 subnets
O       192.168.10.0 [110/4] via 10.10.10.22, 00:38:21, GigabitEthernet1/0/1
O       192.168.10.64 [110/4] via 10.10.10.22, 00:38:21, GigabitEthernet1/0/1
O       192.168.10.128 [110/4] via 10.10.10.22, 00:38:21, GigabitEthernet1/0/1
O       192.168.10.192 [110/4] via 10.10.10.22, 00:38:21, GigabitEthernet1/0/1
     192.168.11.0/24 is variably subnetted, 6 subnets, 2 masks
O       192.168.11.0 [110/4] via 10.10.10.22, 00:38:21, GigabitEthernet1/0/1
O       192.168.11.64 [110/4] via 10.10.10.22, 00:38:21, GigabitEthernet1/0/1

LT3_L35_535715#
```

- L3S Lantai 4

```
LT4_L35_535715#show ip route ospf
     10.0.0.0/8 is variably subnetted, 13 subnets, 2 masks
O       10.10.10.0 [110/65] via 10.10.10.46, 00:02:16, GigabitEthernet1/0/2
O       10.10.10.8 [110/65] via 10.10.10.26, 00:02:16, GigabitEthernet1/0/1
O       10.10.10.12 [110/66] via 10.10.10.26, 00:02:16, GigabitEthernet1/0/1
                    [110/66] via 10.10.10.46, 00:02:16, GigabitEthernet1/0/2
O       10.10.10.16 [110/3] via 10.10.10.26, 00:02:16, GigabitEthernet1/0/1
O       10.10.10.28 [110/3] via 10.10.10.46, 00:02:16, GigabitEthernet1/0/2
O       10.10.10.32 [110/3] via 10.10.10.46, 00:02:16, GigabitEthernet1/0/2
O       10.10.10.36 [110/3] via 10.10.10.26, 00:02:16, GigabitEthernet1/0/1
O       10.10.10.48 [110/2] via 10.10.10.26, 00:02:16, GigabitEthernet1/0/1
O       10.10.10.52 [110/2] via 10.10.10.46, 00:02:16, GigabitEthernet1/0/2
     192.168.10.0/26 is subnetted, 4 subnets
O       192.168.10.0 [110/4] via 10.10.10.46, 00:02:16, GigabitEthernet1/0/2
O       192.168.10.64 [110/4] via 10.10.10.46, 00:02:16, GigabitEthernet1/0/2
O       192.168.10.128 [110/4] via 10.10.10.46, 00:02:16, GigabitEthernet1/0/2
O       192.168.10.192 [110/4] via 10.10.10.46, 00:02:16, GigabitEthernet1/0/2
     192.168.11.0/24 is variably subnetted, 6 subnets, 2 masks
O       192.168.11.0 [110/4] via 10.10.10.46, 00:02:16, GigabitEthernet1/0/2
O       192.168.11.64 [110/4] via 10.10.10.46, 00:02:16, GigabitEthernet1/0/2

LT4 L35 535715#
```

- RT 1

```
LT1_RT_535715#show ip route ospf
     10.0.0.0/8 is variably subnetted, 18 subnets, 2 masks
O       10.10.10.8 [110/65] via 10.10.10.50, 01:09:14, GigabitEthernet0/0
O       10.10.10.20 [110/4] via 10.10.10.50, 00:03:32, GigabitEthernet0/0
O       10.10.10.24 [110/2] via 10.10.10.50, 01:09:14, GigabitEthernet0/0
O       10.10.10.28 [110/5] via 10.10.10.50, 01:09:14, GigabitEthernet0/0
O       10.10.10.32 [110/5] via 10.10.10.50, 00:29:43, GigabitEthernet0/0
O       10.10.10.40 [110/2] via 10.10.10.50, 01:09:14, GigabitEthernet0/0
O       10.10.10.44 [110/3] via 10.10.10.50, 00:03:32, GigabitEthernet0/0
O       10.10.10.52 [110/4] via 10.10.10.50, 01:09:14, GigabitEthernet0/0
     192.168.10.0/26 is subnetted, 4 subnets
O       192.168.10.0 [110/6] via 10.10.10.50, 01:09:14, GigabitEthernet0/0
O       192.168.10.64 [110/6] via 10.10.10.50, 01:09:14, GigabitEthernet0/0
O       192.168.10.128 [110/6] via 10.10.10.50, 01:09:14, GigabitEthernet0/0
O       192.168.10.192 [110/6] via 10.10.10.50, 01:09:14, GigabitEthernet0/0
     192.168.11.0/26 is subnetted, 4 subnets
O       192.168.11.0 [110/6] via 10.10.10.50, 01:09:14, GigabitEthernet0/0
O       192.168.11.64 [110/6] via 10.10.10.50, 01:09:14, GigabitEthernet0/0
O       192.168.11.128 [110/3] via 10.10.10.50, 00:03:32, GigabitEthernet0/0
O       192.168.11.192 [110/3] via 10.10.10.50, 00:03:32, GigabitEthernet0/0
     192.168.12.0/26 is subnetted, 4 subnets
O       192.168.12.0 [110/3] via 10.10.10.50, 01:09:14, GigabitEthernet0/0
O       192.168.12.64 [110/3] via 10.10.10.50, 00:03:32, GigabitEthernet0/0
O       192.168.12.128 [110/3] via 10.10.10.50, 01:09:14, GigabitEthernet0/0
O       192.168.12.192 [110/3] via 10.10.10.50, 01:09:14, GigabitEthernet0/0

LT1_RT_535715#
```

- RT 2

```
LT2_RT_535715#show ip route ospf
     10.0.0.0/8 is variably subnetted, 18 subnets, 2 masks
O       10.10.10.0 [110/65] via 10.10.10.54, 01:09:58, GigabitEthernet0/0
O       10.10.10.16 [110/2] via 10.10.10.33, 00:30:22, GigabitEthernet0/2
O       10.10.10.20 [110/2] via 10.10.10.54, 01:09:58, GigabitEthernet0/0
O       10.10.10.24 [110/3] via 10.10.10.54, 00:04:11, GigabitEthernet0/0
O       10.10.10.36 [110/2] via 10.10.10.29, 01:09:58, GigabitEthernet0/1
O       10.10.10.40 [110/4] via 10.10.10.54, 00:04:11, GigabitEthernet0/0
O       10.10.10.44 [110/2] via 10.10.10.54, 01:09:58, GigabitEthernet0/0
O       10.10.10.48 [110/4] via 10.10.10.54, 01:09:58, GigabitEthernet0/0
     192.168.10.0/26 is subnetted, 4 subnets
O       192.168.10.0 [110/2] via 10.10.10.29, 01:09:58, GigabitEthernet0/1
O       192.168.10.64 [110/2] via 10.10.10.29, 01:09:58, GigabitEthernet0/1
O       192.168.10.128 [110/2] via 10.10.10.29, 01:09:58, GigabitEthernet0/1
O       192.168.10.192 [110/2] via 10.10.10.29, 01:09:58, GigabitEthernet0/1
     192.168.11.0/26 is subnetted, 4 subnets
O       192.168.11.0 [110/2] via 10.10.10.29, 01:09:58, GigabitEthernet0/1
O       192.168.11.64 [110/2] via 10.10.10.29, 01:09:58, GigabitEthernet0/1
O       192.168.11.128 [110/3] via 10.10.10.54, 00:04:11, GigabitEthernet0/0
O       192.168.11.192 [110/3] via 10.10.10.54, 00:04:11, GigabitEthernet0/0
     192.168.12.0/26 is subnetted, 4 subnets
O       192.168.12.0 [110/3] via 10.10.10.54, 01:09:58, GigabitEthernet0/0
O       192.168.12.64 [110/3] via 10.10.10.54, 00:04:11, GigabitEthernet0/0
O       192.168.12.128 [110/3] via 10.10.10.54, 01:09:58, GigabitEthernet0/0
O       192.168.12.192 [110/3] via 10.10.10.54, 01:09:58, GigabitEthernet0/0
```

- RT 3

```
LT3_RT_535715#show ip route ospf
     10.0.0.0/8 is variably subnetted, 17 subnets, 2 masks
O       10.10.10.8 [110/65] via 10.10.10.53, 01:05:08, GigabitEthernet0/0
O       10.10.10.12 [110/65] via 10.10.10.53, 01:05:08, GigabitEthernet0/0
O       10.10.10.16 [110/3] via 10.10.10.53, 00:31:01, GigabitEthernet0/0
O       10.10.10.24 [110/2] via 10.10.10.45, 00:04:59, GigabitEthernet0/2
O       10.10.10.28 [110/2] via 10.10.10.53, 01:05:08, GigabitEthernet0/0
O       10.10.10.32 [110/2] via 10.10.10.53, 00:31:11, GigabitEthernet0/0
O       10.10.10.36 [110/3] via 10.10.10.53, 01:05:08, GigabitEthernet0/0
O       10.10.10.40 [110/3] via 10.10.10.45, 00:04:59, GigabitEthernet0/2
O       10.10.10.48 [110/3] via 10.10.10.45, 00:04:59, GigabitEthernet0/2
     192.168.10.0/26 is subnetted, 4 subnets
O       192.168.10.0 [110/3] via 10.10.10.53, 01:05:08, GigabitEthernet0/0
O       192.168.10.64 [110/3] via 10.10.10.53, 01:05:08, GigabitEthernet0/0
O       192.168.10.128 [110/3] via 10.10.10.53, 01:05:08, GigabitEthernet0/0
O       192.168.10.192 [110/3] via 10.10.10.53, 01:05:08, GigabitEthernet0/0
     192.168.11.0/26 is subnetted, 4 subnets
O       192.168.11.0 [110/3] via 10.10.10.53, 01:05:08, GigabitEthernet0/0
O       192.168.11.64 [110/3] via 10.10.10.53, 01:05:08, GigabitEthernet0/0
O       192.168.11.128 [110/2] via 10.10.10.45, 00:04:59, GigabitEthernet0/2
O       192.168.11.192 [110/2] via 10.10.10.45, 00:04:59, GigabitEthernet0/2
     192.168.12.0/26 is subnetted, 4 subnets
O       192.168.12.0 [110/2] via 10.10.10.45, 00:04:59, GigabitEthernet0/2
O       192.168.12.64 [110/2] via 10.10.10.45, 00:04:59, GigabitEthernet0/2
O       192.168.12.128 [110/2] via 10.10.10.45, 00:04:59, GigabitEthernet0/2
O       192.168.12.192 [110/2] via 10.10.10.45, 00:04:59, GigabitEthernet0/2

[illegible]
```

- RT 4

```
LT4_RT_535715#show ip route ospf
     10.0.0.0/8 is variably subnetted, 17 subnets, 2 masks
O       10.10.10.0 [110/65] via 10.10.10.49, 01:11:27, GigabitEthernet0/0
O       10.10.10.12 [110/65] via 10.10.10.49, 01:11:27, GigabitEthernet0/0
O       10.10.10.16 [110/2] via 10.10.10.49, 01:11:27, GigabitEthernet0/0
O       10.10.10.20 [110/3] via 10.10.10.25, 00:05:40, GigabitEthernet0/1
O       10.10.10.28 [110/4] via 10.10.10.25, 00:05:40, GigabitEthernet0/1
O       10.10.10.32 [110/4] via 10.10.10.25, 00:05:40, GigabitEthernet0/1
O       10.10.10.36 [110/2] via 10.10.10.49, 01:11:27, GigabitEthernet0/0
O       10.10.10.44 [110/2] via 10.10.10.25, 00:05:40, GigabitEthernet0/1
O       10.10.10.52 [110/3] via 10.10.10.25, 00:05:40, GigabitEthernet0/1
     192.168.10.0/26 is subnetted, 4 subnets
O       192.168.10.0 [110/5] via 10.10.10.25, 00:05:40, GigabitEthernet0/1
O       192.168.10.64 [110/5] via 10.10.10.25, 00:05:40, GigabitEthernet0/1
O       192.168.10.128 [110/5] via 10.10.10.25, 00:05:40, GigabitEthernet0/1
O       192.168.10.192 [110/5] via 10.10.10.25, 00:05:40, GigabitEthernet0/1
     192.168.11.0/26 is subnetted, 4 subnets
O       192.168.11.0 [110/5] via 10.10.10.25, 00:05:40, GigabitEthernet0/1
O       192.168.11.64 [110/5] via 10.10.10.25, 00:05:40, GigabitEthernet0/1
O       192.168.11.128 [110/2] via 10.10.10.25, 00:05:40, GigabitEthernet0/1
O       192.168.11.192 [110/2] via 10.10.10.25, 00:05:40, GigabitEthernet0/1
     192.168.12.0/26 is subnetted, 4 subnets
O       192.168.12.0 [110/2] via 10.10.10.25, 00:05:40, GigabitEthernet0/1
O       192.168.12.64 [110/2] via 10.10.10.25, 00:05:40, GigabitEthernet0/1
O       192.168.12.128 [110/2] via 10.10.10.25, 00:05:40, GigabitEthernet0/1
O       192.168.12.192 [110/2] via 10.10.10.25, 00:05:40, GigabitEthernet0/1

LT4 RT 535715#
```

3. Cek koneksi. Ambil salah satu contoh PING antar perangkat.

```
C:\>ping 192.168.11.2

Pinging 192.168.11.2 with 32 bytes of data:

Reply from 192.168.11.2: bytes=32 time=38ms TTL=127
Reply from 192.168.11.2: bytes=32 time=1ms TTL=127
Reply from 192.168.11.2: bytes=32 time=10ms TTL=127
Reply from 192.168.11.2: bytes=32 time<1ms TTL=127

Ping statistics for 192.168.11.2:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 38ms, Average = 12ms
```

PC pada departemen Management dapat PING ke PC pada departemen Accounts

4. Cek Koneksi Perangkat Wireless. Ujicoba dengan PING ke PC di departemen lain

```
C:\>ping 192.168.12.3

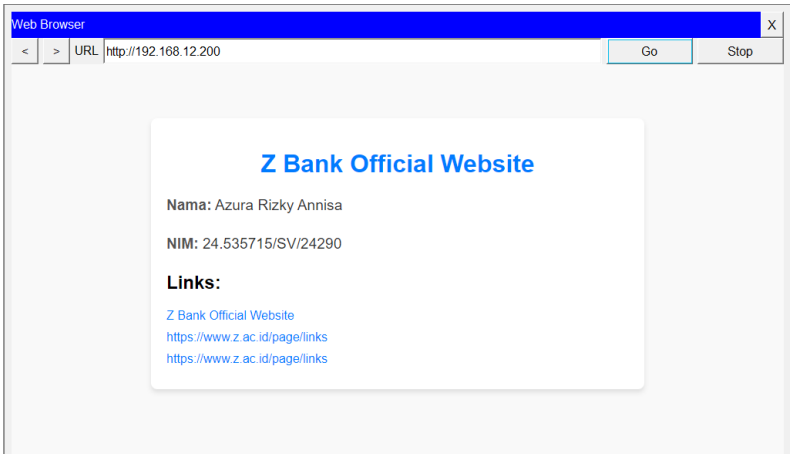
Pinging 192.168.12.3 with 32 bytes of data:

Reply from 192.168.12.3: bytes=32 time=31ms TTL=128
Reply from 192.168.12.3: bytes=32 time=92ms TTL=128
Reply from 192.168.12.3: bytes=32 time=24ms TTL=128
Reply from 192.168.12.3: bytes=32 time=10ms TTL=128

Ping statistics for 192.168.12.3:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 10ms, Maximum = 92ms, Average = 39ms
```

Laptop ping departemen Customer

5. Ujicoba website. Akses alamat web server pada salah satu perangkat.



PEMBAHASAN

Praktikum perancangan jaringan ini bertujuan untuk memahami dan mengimplementasikan desain jaringan berhirarki berbasis pendekatan top-down, yang mencakup konfigurasi pada lapisan akses (access layer), distribusi (distribution layer), hingga inti (core layer). Pada tahap awal, dilakukan konfigurasi dasar pada seluruh perangkat jaringan seperti router dan switch, meliputi penamaan hostname sesuai identitas perangkat, serta pengaturan akses untuk memastikan pengelolaan perangkat dapat dilakukan dengan baik. Tahapan ini menjadi fondasi penting sebelum masuk ke konfigurasi yang lebih kompleks seperti VLAN, routing, dan layanan jaringan.

Pada lapisan akses, dilakukan segmentasi jaringan menggunakan Virtual Local Area Network (VLAN) untuk memisahkan jaringan berdasarkan departemen, seperti Management, Research, HR, Marketing, Accounts, Finance, Logistic, Customer Care, Guest Area, Administrasi, ICT, dan Server. Setiap VLAN dikonfigurasikan pada access switch dan dihubungkan ke perangkat end-user melalui access port, sedangkan koneksi ke multilayer switch menggunakan trunk port agar dapat membawa berbagai VLAN sekaligus. Hasil pengujian menggunakan perintah show vlan brief menunjukkan bahwa setiap VLAN telah terbentuk dengan benar dan port telah terasosiasi sesuai dengan fungsi departemennya.

Pada lapisan distribusi, digunakan perangkat Multilayer Switch (Layer 3 Switch) untuk menangani inter-VLAN routing. Setiap VLAN dibuatkan Switch Virtual Interface (SVI) yang diberikan alamat IP sebagai default gateway bagi host dalam VLAN tersebut. Berdasarkan hasil pengujian menggunakan perintah show ip interface brief, seluruh interface VLAN berada dalam kondisi up/up, yang menunjukkan bahwa inter-VLAN routing telah berjalan dengan baik. Selain itu, konfigurasi ip helper-address diterapkan pada setiap SVI untuk memungkinkan komunikasi antara client dengan DHCP server yang berada di jaringan berbeda.

Sistem pengalamatan IP pada jaringan ini menggunakan mekanisme dinamis melalui DHCP Server yang ditempatkan pada segmen server. Karena DHCP menggunakan broadcast yang tidak dapat melewati batas VLAN, maka multilayer switch berperan sebagai DHCP relay agent untuk meneruskan permintaan dari client ke server. Hasil pengujian menunjukkan bahwa perangkat client seperti PC dan laptop berhasil memperoleh IP address secara otomatis sesuai dengan subnet masing-masing, lengkap dengan gateway dan parameter jaringan lainnya.

Untuk mendukung konektivitas antar jaringan yang lebih luas, diterapkan protokol routing dinamis Open Shortest Path First (OSPF) pada seluruh router dan multilayer switch. OSPF dikonfigurasi pada area backbone (area 0) dengan mendeklarasikan network yang terhubung langsung pada masing-masing perangkat. Selama proses implementasi, ditemukan kendala berupa status neighbor OSPF yang berada pada kondisi EXSTART, yang mengindikasikan ketidakstabilan adjacency. Permasalahan ini diselesaikan dengan mengoptimalkan konfigurasi OSPF menggunakan metode passive-interface pada interface VLAN, sehingga adjacency hanya terjadi pada link backbone. Setelah perbaikan, status neighbor berubah menjadi FULL, dan tabel routing berhasil terbentuk sebagaimana terlihat pada perintah show ip route.

Tahap akhir melibatkan pengujian konektivitas dan layanan jaringan. Pengujian dilakukan dengan metode ping antar perangkat yang berbeda VLAN maupun berbeda lantai, yang menunjukkan bahwa komunikasi antar jaringan telah berjalan dengan baik setelah routing OSPF stabil. Selain itu, pengujian layanan web dilakukan dengan mengakses alamat IP server melalui browser pada client, yang berhasil menampilkan halaman web yang telah dikonfigurasi sebelumnya. Hal ini menandakan bahwa seluruh komponen jaringan, mulai dari VLAN, routing, DHCP, hingga layanan aplikasi, telah terintegrasi dan berfungsi sesuai dengan perancangan.

Secara keseluruhan, praktikum ini berhasil mengimplementasikan jaringan berbasis arsitektur hirarki yang terstruktur dan efisien. Setiap lapisan jaringan menjalankan fungsinya masing-masing dengan baik, mulai dari segmentasi di access layer, routing di distribution layer, hingga konektivitas antar jaringan di core layer. Proses troubleshooting yang

dilakukan selama praktikum juga memberikan pemahaman lebih dalam terkait dinamika konfigurasi jaringan, khususnya dalam menangani permasalahan routing dan komunikasi antar perangkat.

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